

VIKRAM SHARMA

QUENCHED: Children at a water treatment plant in Gidderbaha



Reversing Impurity

How a seemingly expensive technology was used to rid 42 villages of the old plague of water pollution

■ By Ramesh Vinayak

Manjit Kaur could hardly have believed it but she has been cured of a two-decade-old list of ailments. Her cure has come neither through a doctor nor a stack of pills, but instead from a community-based safe drinking water project commissioned in her native Muksar district, Punjab, four months ago. Now Kaur, 45, is among the first in queue to fetch a 20-litre can of treated water in her village twice a day. It was contaminated water that gave her joint pains and bleeding gums. Now that the water is clean, Kaur's health is reviving.

The project, Shudh Jal Pariyojna, was launched by Hyderabad-based Naandi Foundation and is running in 42 villages in the Gidderbaha subdivision. It centres around the reverse osmosis (RO) technology provided by the Tata Project Limited. The scheme aims to turn hard groundwater into potable water. While the state Government contributed Rs 4 crore, the local panchayats provided a plot and an electricity connection for the treatment centres.

Gidderbaha's natives had been consuming unhygienic drinking water laced with abnormally high total dissolved solids (TDS). Its TDS count ranges from 1,500 to 9,000—when the prescribed standards set by WHO for safe drinking water are 50 to 70 TDS. The water here comprises inorganic salts like calcium, magnesium, potassium, sodium, bicarbonates,



chlorides and sulfates. These minerals are the cause of several water-borne diseases, including cancer. Worse yet, the Government's drinking water schemes run on canal water, which is sourced from the Sutlej river and found to be toxic by several state agencies. They use an outdated sand-filter method, which doesn't clean the water proficiently.

To urban users, RO is considered among the more expensive water filtration technologies. At the community level, this scheme has ensured that a litre of treated water is sold at 10 paise for domestic use and 15 paise for commercial use. More than half the subdivision's 3.5 lakh population has subscribed to the scheme and the remaining 13 villages should have RO centres within two months. Two weeks after the launch, outlets doubled in Gidderbaha town to cope with rising demand. According to state Finance Minister Manpreet Singh Badal, the driving force behind the project, it has not only brought tangible health benefits for the locals but has also revealed that "people are willing to pay for quality public service".

In an ingenious way to bolster awareness, the foundation supplies all government schools with highly-subsidised RO-treated water. "We want to make women and children the agents of change," says Subdivisional Magistrate Rishi Pal Singh. A home-delivery system will soon be introduced and the scheme will be put into effect in Bhatinda and Mansa. ■